6. SPECTRUMCHAIN - Decentralized Agricultural Monitoring Network

This project utilizes a hyperspectral imaging constellation to monitor global crop health while ensuring farmers maintain ownership of their field data through NFTs. Unlike traditional expensive systems, this blockchain-based marketplace allows farmers to access AI-driven insights regarding disease detection and water stress while participating in a fair data economy.

Mission Brief

Design a hyperspectral imaging satellite constellation that monitors crop health globally and creates a blockchain-based data marketplace where farmers access AI-driven insights (disease detection, water stress, yield prediction) while retaining ownership of their field data via NFTs. Traditional systems are expensive and farmers don't own their data.

Satellite Requirements

- **Orbit:** Sun-synchronous LEO 600 km, 4-8 satellite constellation
- **Coverage:** 3-day global revisit, priority areas (US Midwest, India, Brazil)
- **Payload:** Hyperspectral imager (400-2500 nm, 200+ spectral bands, 30m GSD)
- **Power:** 80W (high-power imager + onboard processing)
- **Mass:** <150 kg (miniaturized hyperspectral payload)
- **Comms:** X-band 200 Mbps (massive data volumes)

Web3 System

Data Flow: Satellite captures hyperspectral image of farmland → Onboard AI calculates vegetation indices (NDVI, NDWI, chlorophyll content) → Raw data + AI outputs stored on IPFS → Each field's data minted as NFT owned by farmer → Farmer grants access to agronomists/insurers via smart contract → AI models train on aggregated data (privacy-preserving) → Farmers earn tokens when their data improves models

Key Smart Contracts:

1. **FieldRegistry** - Maps GPS boundaries to farmer wallet addresses
2. **DataNFT** - Each field's time-series data as NFT (farmer owns, can sell/license)
3. **AccessControl** - Farmers grant granular permissions (insurance can see yield, competitors cannot)
4. **AITrainingMarket** - Researchers pay farmers for anonymized training data
5. **CropInsurance** - Automated payouts based on on-chain vegetation indices (drought = instant payout)

Round 1 constraints:

1. Mission Objectives & Success Criteria

Every design must start with the **"Why."** In a hackathon, your mission objective isn't just a goal; it's a boundary.

- **Operational Lifetime:** How long must the satellite survive? (e.g., 6 months for a tech demo vs. 5 years for a weather satellite).
- **Target Orbit:** This dictates almost everything else.
    - **LEO (Low Earth Orbit):** Lower latency, higher resolution, but high atmospheric drag.
    - **GEO (Geostationary Orbit):** Constant coverage of one area, but extremely high launch costs and radiation.
- **Data Latency:** Does the data need to be "real-time" (for disaster relief) or can it be stored and forwarded once a day?

2. Payload Constraints

The payload is the "customer" of the satellite; everything else (power, thermal, comms) exists just to keep it happy.

- **SWaP-C:** This is the golden rule of satellite design: **S**ize, **W**eight, **a**nd **P**ower - **C**ost.
- **Mass & Volume:** Most hackathons use the "U" standard (1U = 10x10x10 cm). You'll likely be restricted to a 3U or 6U form factor.
- **Duty Cycle:** Can the payload run 24/7, or does it need to shut down to recharge batteries while the satellite is in the Earth's shadow?

3. Budget & Resource Estimation

In a real-world scenario (and a good hackathon), "unlimited money" isn't a thing.

- **Total Cost of Ownership (TCO):** You must account for:
    1. **Development:** Components and labor.
    2. **Launch:** Often calculated per kilogram (e.g., $50,000/kg for LEO).
    3. **Operations:** Ground station fees to talk to your satellite.
- **COTS vs. Custom:** Using **Commercial Off-The-Shelf (COTS)** parts is cheaper and faster but might not be "space-hardened" against radiation compared to custom, radiation-hardened components.

4. Technical Constraints & Requirements

Here is the formal, structured expansion for **SPECTRUMCHAIN**, designed for a professional "Round 1" technical presentation.

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1. Advanced Sensing: The Power of Hyperspectral Imaging

- **Beyond the Visible:** Unlike standard RGB cameras, our **Hyperspectral Imager** captures over 200 spectral bands (400–2500 nm). This allows the satellite to see chemical signatures, such as chlorophyll content, nitrogen levels, and early-stage water stress, long before the human eye can see a crop failing.

- **Precision Agriculture:** With a **30m Ground Sample Distance (GSD)**, the system is calibrated to provide field-level insights. It can differentiate between a healthy crop, a nutrient-deficient one, and one under attack by specific pests.
- **Global Monitoring:** A constellation of 4–8 satellites in a **Sun-synchronous orbit** ensures a 3-day revisit rate for every farm on Earth, providing the consistent time-series data needed for accurate yield prediction.

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2. Data Sovereignty: Your Farm, Your NFT

- **The DataNFT Model:** Traditional ag-tech companies often collect farmer data for free and sell it back to them as expensive reports. **SPECTRUMCHAIN** flips this: every time-series data packet for a field is minted as an **NFT owned by the farmer**.
- **Granular Access Control:** Through **Smart Contracts**, the farmer decides who sees their data. They can grant "View Only" access to an insurance company to prove a drought claim, while keeping their yield secrets hidden from competitors.
- **Privacy-Preserving AI:** Researchers can pay to train AI models on aggregated data, but the **AITrainingMarket** contract ensures farmers are compensated in tokens for their contribution without ever losing "ownership" of the raw source.

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3. Automated Economy & Insurance

- **The "FieldRegistry":** We use a blockchain ledger to map physical GPS boundaries to digital wallet addresses. This creates a secure link between the land and the financial benefits it generates.
- **Instant Drought Payouts:** The **CropInsurance** contract is linked directly to the satellite's vegetation indices (like NDVI/NDWI). If the moisture levels in a specific GPS grid drop below a critical threshold for 10 days, a payout is triggered **automatically** to the farmer's wallet.
- **Low-Friction Marketplace:** By removing agricultural consultants and manual adjusters, we reduce the cost of high-end crop monitoring by up to 80%, making "Space-Age Farming" affordable for smallholder farmers in India and Brazil.

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4. Engineering & Data Management

- **Onboard Edge-AI:** Hyperspectral data is massive. To manage this, the satellite uses **Onboard AI** to calculate indices (like Chlorophyll content) in orbit, only downlinking the high-value "intelligence" packets to save on bandwidth.
- **High-Speed Throughput:** We utilize **X-band 200 Mbps** downlinks to handle the dense spectral data cubes required for deep agricultural analysis.
- **Miniaturized Payload:** By condensing a complex lab-grade spectrometer into a **<150 kg** satellite frame, we significantly lower launch costs while maintaining "Gold Standard" data quality.